Classification Metrics

Homework #2

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Invalid Date

```r
library(tidyverse)
library(skimr)
library(corrplot)
library(ggplot2)
library(GGally)
library(janitor)
```

1 Homework Solutions

1. *Download the classification output data set (attached in Blackboard to the assignment).*

```r
classifications_output_data <- read.csv("classification-output-data (2).csv")

classifications_output_data |> skim()
```

Table 1: Data summary

Name	classifications_output_da...
Number of rows	181
Number of columns	11
Column type frequency:	
numeric	11
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
pregnant	0	1	3.86	3.24	0.00	1.00	3.00	6.00	15.00	
glucose	0	1	118.30	30.48	57.00	99.00	112.00	136.00	197.00	
diastolic	0	1	71.70	11.80	38.00	64.00	70.00	78.00	104.00	
skinfold	0	1	19.80	15.69	0.00	0.00	22.00	32.00	54.00	
insulin	0	1	63.77	88.73	0.00	0.00	0.00	105.00	543.00	
bmi	0	1	31.58	6.66	19.40	26.30	31.60	36.00	50.00	
pedigree	0	1	0.45	0.28	0.09	0.26	0.39	0.58	2.29	
age	0	1	33.31	11.18	21.00	24.00	30.00	41.00	67.00	
class	0	1	0.31	0.47	0.00	0.00	0.00	1.00	1.00	
scored.class	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
scored.probability	0	1	0.30	0.23	0.02	0.12	0.24	0.43	0.95	

2. *The data set has three key columns we will use:*

- `class`: *the actual class for the observation*
- `scored.class`: *the predicted class for the observation (based on a threshold of 0.5)*
- `scored.probability`: *the predicted probability of success for the observation*

  Use the table() function to get the raw confusion matrix for this scored dataset. Make sure you understand the output. In particular, do the rows represent the actual or predicted class? The columns?

```r
confusion_matrix <- table(
classifications_output_data$class,
classifications_output_data$scored.class
)
confusion_matrix
```

```
      0   1
  0 119   5
  1  30  27
```

```r
tn <- confusion_matrix[1, 1]
fn <- confusion_matrix[2, 1]


# print
```

```r
print(paste("True Negatives:", tn))
```

```
[1] "True Negatives: 119"
```

```r
print(paste("False Negatives:", fn))
```

```
[1] "False Negatives: 30"
```

The rows represent the actual class (`class`) and the columns represent the predicted class (`scored.class`). For example, there are 119 observations where the actual class is 0 and the predicted class is also 0 (true negatives), and there are 30 observations where the actual class is 1 but the predicted class is 0 (false negatives).

3. *Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the accuracy of the predictions.*

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

```r
calculate_accuracy <- function(data, actual_col, predicted_col) {
confusion_matrix <- table(data[[actual_col]], data[[predicted_col]])
tp <- confusion_matrix[2, 2]
tn <- confusion_matrix[1, 1]
fp <- confusion_matrix[1, 2]
fn <- confusion_matrix[2, 1]
accuracy <- (tp + tn) / (tp + fp + tn + fn)
return(accuracy)
}

# Test function
accuracy <- calculate_accuracy(classifications_output_data, "class", "scored.class")
print(paste("Accuracy:", accuracy))
```

```
[1] "Accuracy: 0.806629834254144"
```

4. *Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the classification error rate of the predictions.*

$$ClassificationErrorRate = \frac{FP + FN}{TP + FP + TN + FN}$$

Verify that you get an accuracy and an error rate that sums to one.

```r
calculate_error_rate <- function(data, actual_col, predicted_col) {
confusion_matrix <- table(data[[actual_col]], data[[predicted_col]])
tp <- confusion_matrix[2, 2]
tn <- confusion_matrix[1, 1]
fp <- confusion_matrix[1, 2]
fn <- confusion_matrix[2, 1]
error_rate <- (fp + fn) / (tp + fp + tn + fn)
return(error_rate)
}

# Test function
error_rate <- calculate_error_rate(classifications_output_data, "class", "scored.class")
print(paste("Error Rate:", error_rate))
```

```
[1] "Error Rate: 0.193370165745856"
```

```r
# Verify that accuracy and error rate sum to 1
print(paste("Sum of accuracy + error rate:", accuracy + error_rate))
```

```
[1] "Sum of accuracy + error rate: 1"
```

5. *Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the precision of the predictions.*

$$Precision = \frac{TP}{TP + FP}$$

6. *Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the sensitivity of the predictions. Sensitivity is also known as recall.*

$$Sensitivity = \frac{TP}{TP + FN}$$

7. *Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the specificity of the predictions.*

$$Specificity = \frac{TN}{TN + FP}$$

```r
calculate_specificity <- function(data, actual_col, predicted_col) {
confusion_matrix <- table(data[[actual_col]], data[[predicted_col]])
tp <- confusion_matrix[2, 2]
tn <- confusion_matrix[1, 1]
```

```r
fp <- confusion_matrix[1, 2]
fn <- confusion_matrix[2, 1]
specificity <- tn / (tn + fp)
return(specificity)
}

# Test function
specificity <- calculate_specificity(classifications_output_data, "class", "scored.class")
print(paste("Specificity:", specificity))
```

```
[1] "Specificity: 0.959677419354839"
```

8. *Write a function that takes the data set as a dataframe, with actual and predicted classifications identified, and returns the F1 score of the predictions.*

$$F1Score = \frac{2 \times Precision \times Sensitivity}{Precision + Sensitivity}$$

9. *Before we move on, let's consider a question that was asked: What are the bounds on the F1 score? Show that the F1 score will always be between 0 and 1. (Hint: If 0 < < 1 and 0 < < 1 then < .)*

10. *Write a function that generates an ROC curve from a data set with a true classification column (class in our example) and a probability column (scored.probability in our example). Your function should return a list that includes the plot of the ROC curve and a vector that contains the calculated area under the curve (AUC). Note that I recommend using a sequence of thresholds ranging from 0 to 1 at 0.01 intervals.*

11. *Use your created R functions and the provided classification output data set to produce all of the classification metrics discussed above.*

12. *Investigate the caret package. In particular, consider the functions confusionMatrix, sensitivity, and specificity. Apply the functions to the data set. How do the results compare with your own functions?*

13. *Investigate the pROC package. Use it to generate an ROC curve for the data set. How do the results compare with your own functions?*